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### Optical transmitter and optical transmission system

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#### Abstract

An optical transmitter includes: a light emitting part that emits light; and a lens that condenses the light emitted from the light emitting part. Further, the optical transmitter includes a multimode optical fiber that is optically coupled to the lens at one end and on which the light that is condensed by the lens is incident. Furthermore, the coefficient of the correlation between the distribution of the intensity of the light in the light emitting part and the distribution of the intensity of return light to the light emitting part, when the light is emitted from the light emitting part, is set to 0.5 or less.

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## Background/Summary

## CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is based on, and claims priority from the prior Japanese Patent Application No. 2022-102420, filed on Jun. 27, 2022, the entire contents of which are incorporated herein by reference.

## TECHNICAL FIELD

(2) The present disclosure relates to an optical transmitter and an optical transmission system.

## BACKGROUND

(3) Conventionally, an optical transmission system has been known in which an electrical signal is converted into an optical signal by a laser provided in an optical transmitter, and the optical signal is transmitted via an optical fiber to a photodiode that is provided in an optical receiver, and is converted into an electrical signal, thereby transmitting a signal.

(4) In recent years, in a system that optically transmits digital signals, the speed of signals that are transmitted has been increasing (for example, transmission at 10 Gbit/s or faster), and thus it is desirable to achieve faster signal transmission more stably by achieving high-quality signal transmission with reduced noise. Further, a system that transmits radio frequency (RF) signals, such as radio on fiber (RoF), is more susceptible to noise, and thus it is desirable to reduce noise in an optical transmission system as much as possible.

(5) One type of noise that affects the signal quality of such an optical transmission system is the effect on the noise characteristics (RIN: relative intensity noise) caused by return light to the laser side. For this reason, reducing the return light to the laser side makes it possible to improve the signal quality of an optical transmission system.

(6) Therefore, as disclosed in JP 2021-028680 A, a configuration has been proposed in which an optical isolator is used to reduce the return light to the laser side.

## SUMMARY OF THE INVENTION

(7) However, if the return light is reduced by using an optical isolator as in the above prior technology, the optical transmission system increases in size and cost.

(8) An object of the disclosure is to provide an optical transmitter and an optical transmission system that are capable of achieving a reduction in size and cost and of suppressing degradation of signal quality due to noise.

(9) An optical transmitter according to an embodiment includes: a light emitting part that emits light; a lens that condenses the light emitted from the light emitting part; and a multimode optical fiber that is optically coupled to the lens at one end and on which the light that is condensed by the lens is incident. A coefficient of a correlation between a distribution of an intensity of the light in the light emitting part and a distribution of an intensity of return light to the light emitting part, when the light is emitted from the light emitting part, is set to 0.5 or less.

(10) An optical system according to an embodiment includes: the optical transmitter; an optical receiver that receives light that is transmitted from the optical transmitter; and an optical fiber that transmits the light that is transmitted from the optical transmitter to the optical receiver.

(11) The above configuration makes it possible to provide an optical transmitter and an optical transmission system that are capable of achieving a reduction in size and cost and of suppressing degradation of signal quality due to noise.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram schematically illustrating an example of an optical transmission system.

(2) FIG. 2 illustrates an example of the optical transmission system by dividing the optical transmission system into an optical transmitter, an optical fiber, and an optical receiver.

(3) FIG. 3 is a cross-sectional view schematically illustrating a multimode optical fiber that is

provided in the optical transmitter.

(4) FIG. 4 is a diagram schematically illustrating an example of a coupling method between a light emitting part and the multimode optical fiber that are provided in the optical transmitter.

(5) FIG. 5 is a diagram schematically illustrating another example of a coupling method between the light emitting part and the multimode optical fiber that are provided in the optical transmitter.

(6) FIG. 6 is a diagram schematically illustrating a modified example of an optical transmission system.

(7) FIG. 7 is a diagram schematically illustrating the distribution of the intensity of light in the light emitting part when light is emitted from the light emitting part and the distribution of the intensity of return light to the light emitting part.

(8) FIG. 8 is a diagram illustrating the distribution of the intensity of light in the light emitting part when light is emitted from the light emitting part, and the distribution of the intensity of return light to the light emitting part, in a normally distributed state.

(9) FIG. 8 is a diagram illustrating the distribution of the intensity of light in the light emitting part when light is emitted from the light emitting part, and the distribution of the intensity of return light to the light emitting part, in a normalized distribution state.

(10) FIG. 9 is a diagram schematically illustrating a microbend part of the multimode optical fiber.

(11) FIG. 10 is a diagram schematically illustrating another modified example of an optical transmission system.

#### DETAILED DESCRIPTION OF THE INVENTION

(12) Various embodiments will be described hereinafter with reference to the accompanying drawings.

(13) Hereafter, an optical transmitter and an optical transmission system according to the present embodiment will be described in detail with reference to the drawings. Note that the dimensional ratios in the drawings are exaggerated for the sake of the explanation and may differ from the actual ratios.

(14) As illustrated in FIGS. 1 and 2, an optical transmission system 1 according to the present embodiment includes: an optical transmitter 10 that is capable of transmitting an optical signal (light) S2; and an optical receiver 20 that is capable of receiving the optical signal S2 transmitted from the optical transmitter 10.

(15) In addition, the optical transmission system 1 includes an optical fiber 30 that is optically coupled to the optical transmitter 10 at one end and optically coupled to the optical receiver 20 at the other end, and transmits the optical signal S2 that is transmitted from the optical transmitter 10 to the optical receiver 20.

(16) The optical transmitter 10 includes a laser (light emitting part) 14 that converts an electrical signal S1 into the optical signal (light) S2, and a drive circuit 13 that drives the laser 14. In addition, the optical receiver 20 includes a photodiode (PD) 24 that converts the optical signal (light) S2 into the electrical signal S1, and a drive circuit 23 that drives the photodiode (PD) 24.

(17) The optical transmitter 10 is connected to a network device or the like, and the optical receiver 20 is connected to another network device or the like, and thus signals from the network device on the side of the optical transmitter 10 are transmitted to the network device on the side of the optical receiver 20 via the optical transmission system 1.

(18) Specifically, the electrical signal S1 that is transmitted from the network device or the like on the side of the optical transmitter 10 to the optical transmitter 10 is converted into the optical signal S2 by the laser 14 driven by the drive circuit 13. The optical signal S2 that is converted from the electrical signal S1 is emitted from the laser 14 to be incident on the optical fiber 30, and transmitted to the optical receiver 20 via the optical fiber 30. The optical signal S2 that is transmitted to the optical receiver 20 is converted into the electrical signal S1 by the photodiode (PD) 24 driven by the drive circuit 23, and the converted electrical signal S1 is transmitted to a device such as the network device on the side of the optical transmitter 10.

(19) Accordingly, the optical transmission system **1** transmits signals by converting the electric signal **S1** into the optical signal **S2** by means of the laser **14** of the optical transmitter **10**, and transmitting the optical signal **S2** to the photodiode **24** of the optical receiver **20** via the optical fiber **30** to convert the optical signal **S2** into the electric signal **S1**.

(20) The optical transmission system **1** having such a configuration can be used in a system that transmits digital signals or a system that transmits radio frequency (RF) signals such as radio on fiber (RoF). That is, the optical transmission system **1** according to the present embodiment can be used for various applications such as broadcasting, communication, infrastructure, and EMC (e.g. television/radio broadcast waves, cellular telephones, GPS, Bluetooth, and Wi-Fi).

(21) The optical transmitter **10** is provided with a case **11** in which the laser **14** and the drive circuit **13** described above are housed. In addition, a multimode optical fiber (HMC-Fiber) **16** for transmitting the optical signal **S2** that is emitted from the laser **14** is housed in the case **11**, and an optical connector **17** is attached to the outside of the case **11** in an exposed state. Further, the optical signal **S2** that is transmitted by the multimode optical fiber **16** is transmitted to the optical fiber **30** via the optical connector **17** by coupling the optical fiber **30** to the optical connector **17**.

(22) The multimode optical fiber **16** is an optical fiber having a plurality of modes that propagate light, and includes a core **161** and cladding **162** that is arranged on the outer periphery of the core **161**, as illustrated in FIG. 3. The multimode optical fiber **16** also includes a tension member **163** that is arranged on the outer periphery of the cladding **162**, and a jacket **164** that is arranged on the outer periphery of the tension member **163** (see FIG. 3).

(23) The core **161** has a function of transmitting the optical signal **S2** and is made of quartz glass or plastic. For example, the core **161** has a diameter of 50 micrometers, 62.5 micrometers, 100 micrometers, or the like. The refractive index of the core **161** may be the same from the center to the outer periphery, or may change smoothly from the center to the outer periphery.

(24) The cladding **162** is made of a material with a lower refractive index than the core **161**, and has a function of refracting the optical signal **S2** that is incident into the core **161**. In addition, the tension member **163** has a function of relaxing the tension or the like that is applied to the core **161** when laying the multimode optical fiber **16**. The jacket **164** has a function of protecting the core **161** and the cladding **162**.

(25) The multimode optical fiber **16** having such a configuration can be optically coupled to the laser **14** by using, for example, the configuration illustrated in FIG. 4.

(26) Specifically, the optical transmitter **10** includes a lens **151** that is capable of condensing the optical signal **S1** that is emitted from the laser **14**, and one end **16a** of the multimode optical fiber **16** is optically coupled to the lens **151**. Accordingly, the multimode optical fiber **16** is optically coupled to the laser **14** via the lens **151**, and the optical signal **S1** that is emitted from the laser **14** and condensed by the lens **151** is incident into the core **161**.

(27) In the configuration illustrated in FIG. 4, the drive circuit **13** and the laser **14** are mounted on a substrate **12**. Specifically, the substrate **12** includes a first substrate **121** on which the drive circuit **13** is mounted, a second substrate **122** on which the laser **14** is mounted, and an FPC substrate **123** that connects the first substrate **121** and the second substrate **122** with each other. Thus, the drive circuit **13** and the laser **14** are electrically connected with each other via the first substrate **121**, the FPC substrate **123**, and the second substrate **122**.

(28) Various types of drive circuits **13** may be used. For example, the drive circuit **13** can be formed using a chip IC, or the drive circuit **13** can be formed using a bias tee, which is a combination of a chip coil and a chip inductor.

(29) In addition, various types of lasers **14** may be used. In the configuration illustrated in FIG. 4, a vertical-cavity surface-emitting laser (VCSEL) driven by direct modulation is used as the laser **14**.

(30) The laser **14**, for example, includes an approximately cylindrical body **141**, and an emission part **142** that emits the converted optical signal **S2** is formed in the body **141** such that an emission surface **142a** is positioned on one surface **1411** of the body **141**.

(31) In addition, a lens holding wall **152** is fixed to the second substrate **122** on which the laser **14** is mounted, and the lens **151** is held on the lens holding wall **152** in a state in which the lens **151** is separated from the emission surface **142a** by a predetermined distance and faces the emission surface **142a**.

(32) A coupling wall **153** is formed on the lens holding wall **152**, and the coupling wall **153** is configured to hold one end **16a** of the multimode optical fiber **16** and optically couple the multimode optical fiber **16** to the lens **151**.

(33) Accordingly, the optical signal **S2** that is emitted from the emission surface **142a** of the laser **14** is condensed by the lens **151** and to be incident into the core **161** of the multimode optical fiber **16**.

(34) In addition, the multimode optical fiber **16** may be optically coupled to the laser **14** as in the configuration illustrated in FIG. 5.

(35) In the configuration illustrated in FIG. 5, the drive circuit **13** and the laser **14** are mounted on the single substrate **12**. Further, in the configuration illustrated in FIG. 5, a reflecting mirror **154** and a condenser lens **155** are provided on the lens holding wall **152**, and the condenser lens **155** is held on the lens holding wall **152** in a state in which the condenser lens **155** is separated from the emission surface **142a** by a predetermined distance and faces the emission surface **142a**. Further, the optical signal **S2** that is emitted from the emission surface **142a** is reflected by the reflecting mirror **154** in a state in which the optical signal **S2** is condensed by the condenser lens **155**, and thus the light path is bent by 90 degrees. Further, the lens **151** is held on the lens holding wall **152** in a state in which the optical signal **S2** reflected by the reflecting mirror **154** can pass through, and one end **16a** of the multimode optical fiber **16** is held on the coupling wall **153**. Accordingly, the multimode optical fiber **16** is optically coupled to the lens **151**. In addition, the optical signal **S2** that is emitted from the emission surface **142a** and reflected by the reflecting mirror **154** via the condensing lens **155** is condensed by the lens **151** and to be incident into the core **161** of the multimode optical fiber **16**.

(36) Incidentally, in FIG. 4 and FIG. 5, a vertical-cavity surface-emitting laser driven by direct modulation is used as the laser **14**; however, the present embodiment is not limited thereto and various types of lasers may be used. For example, a distributed feedback laser diode (DFB-LD) may be used as the laser **14**.

(37) As illustrated in FIG. 6, it is also possible to perform modulation using an external modulator **18**, instead of direct modulation performed by the laser **14**. That is, it is also possible to have the optical signal **S2** that is emitted from the emission surface **142a** be transmitted to the external modulator **18** via a waveguide **19**, and to have the optical signal **S2** that is modulated by the external modulator **18** be incident on the core **161** of the multimode optical fiber **16**. Accordingly, the modulation is performed using the external modulator **18**, which allows faster modulation than when the laser **14** performs modulation directly. As a result, the system can have a configuration that is more suitable for long-distance and high-capacity communications.

(38) In the present embodiment, it is possible to suppress the deterioration of signal quality by reducing the return light to the laser **14** side.

(39) Specifically, the return light to the laser **14** is reduced by breaking the correlation between the distribution **D1** of the intensity of the optical signal **S2** emitted from the laser **14** and the distribution **D2** of the intensity of the return light to the laser **14** (worsening the correlation), and by changing the distribution **D2** of the intensity of the return light to the laser **14**.

(40) In the present embodiment, a coefficient **R** of the correlation between the distribution **D1** and the distribution **D2** is set to 0.5 or less (0 or more to 0.5 or less). This makes it possible to further reduce the return light to the laser **14**, and reduce the influence of the return light to the laser **14** side on the noise characteristics (RIN: relative intensity noise).

(41) Here, the coefficient **R** of the correlation between the distribution **D1** and the distribution **D2** can be obtained as follows. First, using the distribution **D1** of the intensity of the optical signal **S2**

that is emitted from the emission surface **142a** of the laser **14**, the distribution of the intensity of the emitted light (optical signal **S2**) for each position is converted to a normal distribution. Similarly, using the distribution **D2** of the intensity of the return light on one surface **1411** of the main body **141** of the laser **14**, the distribution of the intensity of the return light for each position is converted to a normal distribution.

(42) Next, normalization is performed such that the standard deviation of the normal distribution of the emitted light (optical signal **S2**) is set to 1 in a state in which the relationship of the normal distribution of the emitted light (optical signal **S2**) and the return light is maintained. For example, in a case where the normal distribution of the emitted light has a standard deviation of 0.005 mm and the normal deviation of the return light has a standard deviation of 0.01 mm, when the standard deviation of the normal distribution of the emitted light is set to 1, the standard deviation of the normal distribution of the return light becomes 2.

(43) Next, in a state in which the normal distribution of the emitted light is normalized, the Euclidean distance  $d$  is obtained from the difference in intensity at each position  $x$  (the difference between the power  $a_x$  of the emitted light and the power  $b_x$  of the return light) in the normal distribution of the emitted light (optical signal **S2**) and the normal distribution of the return light. Specifically, the Euclidean distance  $d$  is calculated by the following formula.

$$d = \sqrt{\sum (a_{\text{sub}.x} - b_{\text{sub}.x})^2} \quad [\text{Math. 1}]$$

(44) Accordingly, for example, when the normal distribution of the emitted light has a standard deviation of 0.005 mm and when the normal distribution of the return light has a standard deviation of 0.01 mm, the Euclidean distance  $d$  is 1.6.

(45) Thereafter, the correlation between the two distributions is observed by modulating the Euclidean distance  $d$  obtained in this way. Specifically, the coefficient  $R$  of the correlation is obtained by calculating  $1/(1+d)$ . For example, when the normal distribution of the emitted light has a standard deviation of 0.005 mm and when the normal distribution of the return light has a standard deviation of 0.01 mm, the coefficient  $R$  of the correlation is 0.4.

(46) The coefficient  $R$  of this correlation necessarily becomes smaller when the mode coupling coefficient of the multimode optical fiber that is optically coupled to the lens **151** increases.

(47) Therefore, in the present embodiment, the coefficient  $R$  of the correlation is set to 0.5 or less by using the multimode optical fiber **16** which has a mode coupling coefficient higher than that of the ordinary multimode optical fiber.

(48) Specifically, as illustrated in FIG. **9**, the core **161** of the multimode optical fiber **16** is configured to have a microbend part **161a** that is bent with a radius of curvature smaller than the diameter of the core **161**. This increases the mode coupling coefficient of the multimode optical fiber **16**.

(49) There are various methods for forming the microbend part **161a**. For example, the microbend part **161a** can be formed in the core **161** by shrinking the jacket **164** using heat or the like when manufacturing the multimode optical fiber **16**. In addition, the microbend part **161a** can be formed in the core **161** by applying side pressure. In this respect, when the microbend part **161a** is formed in the core **161**, the loss of light may increase, and thus it is preferable to form the microbend part **161a** such that mode coupling is caused but that the loss of light does not cause a problem.

(50) Thus, when using the multimode optical fiber **16** in which microbending has been caused, the mode coupling coefficient of the optical fiber that is used in the optical transmitter **10** is higher than usual. This makes it possible to increase the mode coupling coefficient with a simple configuration, thereby improving the signal quality of the optical transmitter **10**.

(51) Although the microbend part **161a** can be formed in both a quartz glass core **161** and a plastic core **161**, plastic deformation may occur in a plastic core **161**. For this reason, when forming the microbend part **161a** in the core **161**, it is preferable to use a multimode optical fiber **16** having a core **161** that is made of quartz glass.

(52) Further, the core **161** of the multimode optical fiber **16** may be configured to scatter the optical

signal S2, thereby increasing the mode coupling coefficient of the multimode optical fiber **16**.

(53) Specifically, a refractive index fluctuation is caused in the core **161** of the multimode optical fiber **16**, that is, the refractive index in a certain area of the core **161** is shifted from the normal refractive index, and thus the core **161** of the multimode optical fiber **16** is configured to scatter the optical signal S2 passing into the core **161**. Such a configuration can be possible even when the core **161** is made of quartz glass or plastic.

(54) When the light scattering becomes Rayleigh scattering, the loss of light increases. For this reason, it is preferable to change the refractive index fluctuation from submicrons to microns in size, and to cause, for example, Mie scattering in which scattering toward the rear side or lateral side decreases. This makes it possible to increase scattering toward the front side, thereby making it possible to further reduce the loss of light.

(55) Accordingly, when using the multimode optical fiber **16** with increased light scattering, the mode coupling coefficient of the optical fiber that is used in the optical transmitter **10** is higher than usual. This makes it possible to increase the mode coupling coefficient with a simple configuration, thereby improving the signal quality of the optical transmitter **10**.

(56) It is also possible to increase light scattering in a state in which the microbend part **161a** is formed in the core **161** of the multimode optical fiber **16**.

(57) In order to cause mode coupling, the multimode optical fiber **16** is required to have a certain length. For example, the multimode optical fiber **16** needs to have a length of 0.1 m or longer.

(58) For this reason, as illustrated in FIG. **10**, for example, it is possible to adopt an optical transmitter **10** in a pigtail shape in which the multimode optical fiber **16** is taken out of the case **11** and the optical connector **17** is arranged opposite to the laser **14**. In this case, it is preferable for the multimode optical fiber **16** to have a length of 3.0 m or less from the viewpoint of handling.

(59) It is also possible for the multimode optical fiber **16** in a bent state to be arranged in the case **11**. This makes it possible to cause mode coupling more reliably without increasing the case **11** in size, thereby reducing the optical transmitter **10** in size and improving the signal quality of the optical transmitter **10** more reliably.

(60) In order for the multimode optical fiber **16** to have a size that is suitable for the case **11**, it is preferable for the multimode optical fiber **16** to have a length of 0.5 m or less.

(61) The other end **16b** of the multimode optical fiber **16** is optically coupled to the optical connector **17** that is attached in such a way as to be exposed from the case **11**. Accordingly, the optical signal S2 is transmitted from the optical connector **17** to the outside of the optical transmitter **10** via the multimode optical fiber **16** in which the mode coupling coefficient is enhanced.

(62) By forming the optical transmission system **1** with the optical transmitter **10** having a high signal quality, the signal quality of the optical transmission system **1** can be improved without using an optical isolator.

(63) In the present embodiment, as described above, the optical transmission system **1** is formed by connecting the optical receiver **20** to the optical transmitter **10** via the optical fiber **30**.

(64) The optical receiver **20** that is connected to the optical transmitter **10** at this time includes a case **21** in which the photodiode **24** and the drive circuit **23** described above are housed. The photodiode **24** and the drive circuit **23** are electrically connected with each other by being mounted on a substrate (not illustrated).

(65) In addition, an optical connector **25** is attached to the outside of the case **21** in an exposed state. The optical connector **25** is optically coupled to the photodiode **24**, and the optical signal S2 that is transmitted by the optical fiber **30** is transmitted to the photodiode **24** via the optical connector **25** by coupling the optical fiber **30** to the optical connector **25**.

(66) The optical fiber **30** is optically coupled to the optical connector **17** via a first optical connector part **31**, and optically coupled to the optical connector **25** via a second optical connector part **32**.



(67) At this time, a multimode optical fiber having a high mode coupling coefficient may be used as the optical fiber **30** for optically coupling the optical transmitter **10** and the optical receiver **20** with each other in the same manner as the multimode optical fiber **16**. In addition, a general optical fiber (a multimode optical fiber or a single-mode optical fiber that does not increase the mode coupling coefficient) can be used as the optical fiber **30**. That is, any type of optical fiber can be used as an optical fiber that is coupled to the multimode optical fiber **16** that is provided in the optical transmitter **10**.

(68) However, when a multimode optical fiber having a high mode-coupling coefficient is used as the optical fiber **30**, the transmission distance needs to be reduced due to a large optical loss or a large bandwidth.

(69) In contrast, when a general optical fiber is used as the optical fiber **30**, noise can be suppressed and transmission can be performed over a relatively long distance (for example, about 200 m depending on the transmission speed). Since the optical transmission system **1** according to the present embodiment is a transmission system using a multimode optical fiber, the whole system is intended for transmission over a relatively short distance of several meters to several hundred meters.

(70) The optical transmission system **1** according to the present embodiment makes it possible to improve the signal quality in the transmission of digital signals, and has the effect of further improving the signal quality in a system that transmits radio frequency (RF) signals, such as radio on fiber (RoF), which are susceptible to noise.

(71) It is preferable that the core diameter of the optical fiber **30** be approximately equal to that of the multimode optical fiber **16**.

(72) [Function and Effect]

(73) In the following description, the characteristic configuration of the optical transmitter and the optical transmission system described in the above embodiment and in the modified example thereof, and the effect obtained therefrom, will be described.

(74) The optical transmitter **10** illustrated in the above embodiment and in the modified example thereof includes: the laser (light emitting part) **14** that emits the optical signal (light) **S2**; and the lens **151** that condenses the optical signal (light) **S2** emitted from the laser (light emitting part) **14**. In addition, the optical transmitter **10** includes the multimode optical fiber **16** that is optically coupled to the lens **151** at one end **16a** and on which the optical signal (light) **S2** that is condensed by the lens **151** is incident.

(75) Further, the coefficient **R** of the correlation between the distribution **D1** of the intensity of the optical signal (light) **S2** in the laser (light emitting part) **14** and the distribution **D2** of the intensity of the return light to the laser (light emitting part) **14**, when the optical signal (light) **S2** is emitted from the laser (light emitting part) **14**, is set to 0.5 or less.

(76) In addition, the optical transmission system **1** illustrated in the above embodiment and in the modified example thereof includes: the above optical transmitter **10**; the optical receiver **20** that receives the optical signal **S2** that is transmitted from the optical transmitter **10**; and the optical fiber **30** that transmits the optical signal **S2** that is transmitted from the optical transmitter **10**, to the optical receiver **20**.

(77) Accordingly, by breaking the correlation between the distribution **D1** of the intensity of the optical signal **S2** emitted from the laser **14** and the distribution **D2** of the intensity of the return light to the laser **14**, the distribution **D2** of the intensity of the return light to the laser **14** changes, thereby making it possible to reduce the return light to the laser **14** to some extent. At this time, when the coefficient **R** of the correlation between the distribution **D1** and the distribution **D2** is set to 0.5 or less, the return light to the laser **14** can be further reduced. As a result, the influence of the return light to the laser **14** side on the noise characteristics (RIN: relative intensity noise) can be reduced. For this reason, the signal quality of the optical transmission system **1** can be improved without using an optical isolator.

(78) Therefore, the configuration illustrated in the above embodiment and in the modified example thereof makes it possible to acquire the optical transmitter **10** and the optical transmission system **1** being capable of achieving a reduction in size and cost and of having a high signal quality.

(79) Further, the optical transmitter **10** having a high signal quality makes it possible to reduce noise even when the light is excited in a predetermined mode. For this reason, even when a laser having many low-order modes (a laser with much return light) is used as the laser **14**, it is possible to achieve return light that is equivalent to that of a laser with many high-order modes (a laser with little return light).

(80) When the optical transmitter **10** having a high signal quality is used, it is possible to achieve the optical transmission system **1** having a high signal quality even in a system that transmits radio frequency (RF) signals such radio on fiber (RoF) that are susceptible to noise.

(81) In addition, the optical transmitter **10** may include the case **11** in which the multimode optical fiber **16** is housed, and the optical connector **17** that is attached to the case **11** and connected to the optical fiber **30** outside the case **11**. Further, the multimode optical fiber **16** may be optically coupled to the optical connector **17** at the other end **16b** in a state in which the multimode optical fiber **16** is arranged bent in the case **11**.

(82) For example, when the coefficient  $R$  of the correlation between the distribution **D1** and the distribution **D2** is set to 0.5 or less by using the multimode optical fiber **16** having a higher mode coupling coefficient than usual, the multimode optical fiber **16** needs to be increased in length to some extent in order to cause mode coupling. At this time, when the multimode optical fiber **16** is bent to be arranged in the case **11**, mode coupling can be caused more reliably without increasing the case **11** in size. As a result, it is possible to realize the optical transmitter **10** that is capable of achieving a reduction in size and of having a high signal quality more reliably.

(83) In addition, the core **161** of the multimode optical fiber **16** may have the microbend part **161a** that is bent with a radius of curvature smaller than the diameter of the core **161**.

(84) Accordingly, the mode coupling coefficient of the optical fiber that is used in the optical transmitter **10** is higher than usual by simply replacing the optical fiber used in the optical transmitter **10** with the multimode optical fiber **16** in which microbending has been caused. That is, the coefficient  $R$  of the correlation between the distribution **D1** and the distribution **D2** in the optical transmitter **10** is set to 0.5 or less by simply replacing the optical fiber that is used in the optical transmitter **10** with the multimode optical fiber **16** in which microbending has been caused. This makes it possible to increase the mode coupling coefficient with a simple configuration, thereby improving the signal quality of the optical transmitter **10**.

(85) In addition, the core **161** of the multimode optical fiber **16** may be configured to scatter the optical signal (light) **S2**.

(86) Accordingly, the mode coupling coefficient of the optical fiber that is used in the optical transmitter **10** is higher than usual simply by forming a configuration that causes light scattering in the core **161** of the multimode optical fiber **16**. That is, the coefficient  $R$  of the correlation between the distribution **D1** and the distribution **D2** in the optical transmitter **10** is set to 0.5 or less simply by forming a configuration that causes light scattering in the core **161** of the multimode optical fiber **16**. This makes it possible to increase the mode coupling coefficient with a simple configuration, thereby improving the signal quality of the optical transmitter **10**.

(87) [Other Matters]

(88) Although the present embodiment has been described as above, the present embodiment is not limited to these configurations and various modifications are possible within the scope of the gist of the present embodiment.

(89) For example, the configurations described in the above embodiment and in the modified example thereof may be combined as appropriate.

(90) In the above embodiment and in the modified example thereof, a description has been given regarding an example in which the core diameter of the multimode optical fiber **16** is equal to the

core diameter of the optical fiber 30. However, the core diameter of the multimode optical fiber 16 is not necessarily equal to the core diameter of the optical fiber 30, and the core diameter of the multimode optical fiber 16 may be different from the core diameter of the optical fiber 30.

(91) In addition, although one optical fiber 30 is used as an example in the above embodiment and in the modified example thereof, the number of optical fibers used in the optical transmission system 1 is not limited to one. For example, a plurality of optical fibers that are connected with an optical connector may be used.

(92) In addition, the specifications (shape, size, layout, etc.) of the multimode optical fiber, light emitting part, and other particulars may also be changed as appropriate.

(93) While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

## Claims

1. An optical transmitter comprising: a light emitting part that emits light; a lens that condenses the light emitted from the light emitting part; and a multimode optical fiber that is optically coupled to the lens at one end and on which the light that is condensed by the lens is incident, wherein a coefficient of a correlation between a distribution of an intensity of the light in the light emitting part and a distribution of an intensity of return light to the light emitting part, when the light is emitted from the light emitting part, is set to 0.5 or less.
  2. The optical transmitter according to claim 1, further comprising: a case in which the multimode optical fiber is housed; and an optical connector that is attached to the case and connected to an optical fiber outside the case, wherein the multimode optical fiber is optically coupled to the optical connector at another end in a state in which the multimode optical fiber is arranged bent in the case.
  3. The optical transmitter according to claim 1, wherein a core of the multimode optical fiber has a microbend part that is bent with a radius of curvature smaller than a diameter of the core.
  4. The optical transmitter according to claim 1, wherein a core of the multimode optical fiber is configured to scatter the light.
  5. An optical transmission system comprising: an optical transmitter described in claim 1; an optical receiver that receives light that is transmitted from the optical transmitter; and an optical fiber that transmits the light that is transmitted from the optical transmitter, to the optical receiver.
  6. The optical transmitter according to claim 1, wherein the distribution of the intensity of the light in the light emitting part comprises a standard deviation of 0.005 mm, and the distribution of the intensity of the return light to the light emitting part comprises a standard deviation of 0.01 mm.
  7. The optical transmitter according to claim 6, wherein a Euclidean distance, of a difference in the intensity of the light in the light emitting part at each position in the distribution of the intensity of the light in the light emitting part and the distribution of the intensity of the return light, is 1.6.
  8. The optical transmitter according to claim 7, wherein the coefficient is 0.4.
  9. The optical transmitter according to claim 3, wherein the multimode optical fiber further comprises a jacket surrounding the core, and the microbend part is bent with the radius of curvature smaller than the diameter of the core by the jacket being heat shrunk moreso at the microbend part than at other parts of the multimode optical fiber.
  10. The optical transmitter according to claim 6, wherein the core consists of quartz glass.
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